CONSTANT TEMPERATURE & HUMIDITY TEST CHAMBER

Operation Manual — English Translation

Manufacturer: Dongguan Xin Bao Instrument Co., Ltd. (OTS Technik)

Model identified on specification page: XB-OTS-80H-C

Translated from the 39-page scanned Chinese manual supplied by the user.

Translation note: The source is a scanned document. Some controller screenshots, wiring diagrams, small table entries and numerical characters are not sufficiently clear for reliable transcription. Unreadable items are identified rather than guessed. Confirm all settings and electrical values against the equipment nameplate and the manufacturer before operation.

Source page 1 — Cover and manufacturer details

Constant Temperature and Humidity Test Chamber — Operation Manual

Dongguan Xin Bao Instrument Co., Ltd. (OTS Technik)

Address: Third Industrial Zone, Jingshan, Chashan Town, Dongguan City, Guangdong, China.

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Website shown in the scan: www.otschina.cn

Source page 2 — Contents

The contents page is only partly legible. The following entries can be identified:

• Section 7: Setting the Level 1 / Level 2 Over-temperature Protection.

• Section 9: Pre-test Inspection and Confirmation Items.

• Section 10: Test Chamber Fault Analysis and Troubleshooting (approximately source pages 27–30).

• Section 12: Quarterly Maintenance of the Temperature-Cycling Equipment (approximately source pages 34–35).

Other contents entries cannot be read reliably from the scan.

Source page 3 — Company profile

Dongguan Xin Bao Instrument Co., Ltd. (OTS Technik) describes itself as a high-technology enterprise specializing in research into reliability testing technology and the development, manufacture and sale of climate/environmental test equipment.

The company states that it has a production base of approximately 20,000 m² and holds CNAS-related laboratory/third-party testing credentials, ISO 9001:2015 quality-management certification, TUV certification and CE certification. It also states that it has independent export rights.

Products listed include high/low-temperature test chambers, constant-temperature and humidity chambers, thermal-shock chambers, rapid temperature-change chambers, walk-in temperature/humidity chambers, high/low-temperature and low-pressure chambers, thermal-cycle chambers, precision high-temperature ovens, ageing rooms, combined environmental test chambers and tensile/testing machines.

The manual says the equipment is designed with reference to international standards such as military standards, IEC and DIN, and is used in laboratories and industries including aerospace, aviation, military, automotive, metrology, electronics, railways, power, medical and scientific research.

The company’s stated business focus is climate/environment simulation, meeting customer needs and continuous improvement.

Source page 4 — Preface

Thank you for choosing this company’s product. The manufacturer states that it will provide a product of good quality together with reliable after-sales service.

To protect operators and the equipment, read this operation manual carefully before using the instrument and pay close attention to all operating precautions.

The manual covers the instrument’s design principles, reference standards, construction, operating procedures, calibration, maintenance, possible faults and remedies, and electrical diagrams.

Any test standards mentioned in this manual are for reference only. If the customer has questions or objections, consult the applicable standards or technical data.

Special statement

1. This operation manual does not constitute a basis for making claims against the manufacturer.

2. The manufacturer retains the right to interpret this operation manual.

Source page 5 — Specifications and parameters

Machine model: XB-OTS-80H-C.

Temperature range: −70°C to +150°C. Do not operate outside this range; the manufacturer states that resulting problems are not covered by warranty.

Humidity range: approximately 10% RH to 98% RH. Refer to the humidity operating-range chart in the original manual.

Temperature / humidity control accuracy: ±0.5°C / ±1% RH. The stated performance is to be measured after the temperature sensor has stabilized for 30 minutes.

Uniformity: the scan appears to specify temperature uniformity below approximately ±2°C; humidity uniformity is shown as approximately ±3% RH at humidity above 75% RH and ±5% RH at lower humidity. These characters are not fully clear in the scan; verify with the manufacturer.

Internal chamber dimensions: the scan appears to read approximately 400 × 450 × 400 mm (W × H × D); verify on the equipment documentation/nameplate.

The test specimen must not occupy more than two-thirds of the test-area volume.

External dimensions: subject to the actual machine.

Power supply: AC 220 V. Rated power: approximately 4.5 kW. Protective breaker: 2-pole, 32 A.

Uniformity measurement positions and stabilization requirements are described in the original chart; take measurements at the specified positions after 30 minutes of stable operation.

Several figures in the specification chart are blurred. Treat the values marked “approximately” as provisional and confirm them with the manufacturer before installation or acceptance.

Source page 6 — Controller operation — monitoring screen

1. Monitoring screen

The controller monitoring screen displays the physical quantities being monitored, including sensor readings and the test values that have been set.

1.1 Fixed-value stop and run screens

The controller provides a fixed-value stop screen and a fixed-value run screen.

During fixed-value operation, when the displayed value is above the upper limit of the set value, the downward arrow flashes to indicate cooling. When the displayed value is below the lower limit, the upward arrow flashes to indicate heating. When the displayed value is within the set-value tolerance band, the centre horizontal indicator flashes to indicate that the temperature is being maintained.

The original page contains controller screenshots. Individual screen labels and icon callouts are not all legible in the scan.

Source page 7 — Fixed-value operation — detailed screen

Tap the fixed-value operation screen to enter the detailed view (Section 1.1.1). Tap again to return to the standard fixed-value operation screen (Section 1.1).

The detailed view presents controller status and output indications. The legend identifies cooling output and heating output, along with other controller status indicators.

The running-time field starts counting after the test begins. The timed-duration field shows the preset operating time. The set-value field shows the target value.

The controller screen also displays the current date/time and alarm/status information. Exact icon labels are not reliably legible.

Source page 8 — Program stop and program run screens

The controller has a program-stop screen and a program-run screen.

Values entered while a test program is running are the program’s set values. To change a program set value, switch to Program Setting and edit it there.

The title displayed is the title configured for the currently executing program. To change the title, switch to Program Setting.

The screen shows the current program segment/step and the program number in use.

The elapsed-time counter starts when the test is started and counts toward the operating time configured in the program.

Source page 9 — Program run — detailed screen

Tap the program-run screen to enter the detailed view (Section 1.2.1). Tap again to return to the standard program-run screen (Section 1.2).

The detailed screen displays the program title, temperature and humidity values, current segment/step, elapsed/remaining running information and controller output/status indicators.

The screen includes indications for cooling output, heating output and other operating states. Some labels and numerical examples in the screenshot are too blurred to transcribe accurately.

Source page 10 — Fixed-value settings

This section describes the parameters and related auxiliary functions used during fixed-value operation.

Control mode

Select the required control mode on the fixed-value setting screen. The screenshot distinguishes control-mode settings from stop-mode settings.

Stop method / timing method

The manual illustrates different timing conditions, including timing from the start of operation and timing only after the measured temperature reaches the set temperature.

Return to normal temperature

At the end of a test, the controller can be configured to return to normal/ambient temperature. The illustrated options include returning to the configured temperature value or returning to the normal-temperature environment.

The exact wording of some options in the screenshot is unclear. Confirm the selected stop and return-to-normal-temperature mode on the controller before running a test.

Source page 11 — No legible text

No readable manual text was extracted from this scanned page. It appears to be a blank, separator or image-only page.

Source page 12 — Program editing and program-file management

This page describes program editing and file-management controls.

The program editor allows the user to select a program number and edit the associated program data/segments. The visible range for program numbers appears to be 0–100.

The page also describes saving program settings to a file, assigning a file name and exporting the file. Saved program files can subsequently be selected and imported. The controller identifies available files and displays the import result.

The detailed numeric fields and some button labels are unclear in the scan.

Source page 13 — Program preview

The Program Preview screen displays all segment set values configured under the currently selected program number.

The set values are listed sequentially from top to bottom and represented from left to right on the graph to form a curve.

Use this screen to review the programmed temperature/humidity profile before starting the test.

Source page 14 — Program standby and cycle editing

3.2 Program standby

The standby function holds the equipment in a standby condition until the specified entry conditions are met. The example shown refers to a temperature range around 22.5–23.5°C and a humidity range around 21–25% RH; the exact values and conditions in the graphic should be verified on the controller.

3.3 Cycle editing

The Cycle Editing screen is used to configure a repeated/cyclic program sequence. The page contains a controller screenshot with fields for editing the cycle. Individual labels are not legible enough to translate reliably.

Source page 15 — Test title and control settings

3.4 Test title

Select the required program number to edit the titles for all segments belonging to that program number. The title identifies the test/program on the controller.

3.5 Control settings

This page introduces additional control settings associated with program operation. The settings table and its labels are too blurred for a reliable line-by-line translation.

Source page 16 — No legible text

No readable text was recovered from this page. It may be an image-only controller screen or a blank/separator page.

Source page 17 — Historical data / curve recording

The controller provides historical data recording and curve display functions.

A data range can be selected using upper and lower limits. The display can show selected data items, including temperature set values. The chart display can be expanded or hidden using the relevant icon.

Immediate recording: starts recording as soon as the recording function is activated.

Record during operation: starts recording when the test is started.

Some screen labels and the exact data-selection controls are unclear in the scan.

Source page 18 — Data export, import and data table

4.2 Data export

The Data Export function is used to export recorded test data. The screen shows date/time selection and a data-export operation.

The Data Import function imports previously saved data. A start time and end time can be selected; the controller displays the import result.

4.3 Data table

The Data Table screen lists recorded test data with date/time and measured values. The manual also shows a Historical Curve view and a Data Export function.

The sample dates and numerical readings in the screenshot are examples and are not equipment specifications.

Source page 19 — System settings

The System Settings page includes operating-mode and control-mode options.

Power recovery / restart mode

Stop: after power is interrupted and restored, the machine does not restart automatically.

Cold start: after power is interrupted and restored, the machine restarts.

Hot start: after power is interrupted and restored, the machine resumes/restarts operation according to the configured hot-start behavior.

Control mode

The control mode can be set to temperature-and-humidity control or temperature-only control. The scan notes that this cannot be changed while the monitoring screen is in automatic mode.

Source page 20 — Communication settings

The Communication Settings page configures communication between the controller and an external host/computer.

The visible settings include communication protocol/interface, communication port, slave/station address, parity and baud rate.

The slave address is shown as 1 by default, with a maximum of 247.

The protocol shown is MODBUS-RTU. The baud-rate options visible in the scan include 4,800, 9,600 and 19,200 bit/s.

Parity options include no parity and odd parity; the bit/stop configuration is shown in the controller table. Confirm all communication parameters with the connected supervisory system before use.

Some communication-interface labels and bit-format values are blurred. The controller’s actual menu and the host-system configuration take precedence.

Source page 21 — Permission settings and screen saver

3. Permission settings

The controller supports access permissions. When Parameter Permission is ON, parameter values cannot be modified. When Operation Permission is ON, operation/start-stop actions cannot be modified or performed, as configured.

Password permission is also provided. The screen includes password-related settings.

Screen saver

The screen-saver settings include a delay/time setting, screen-saver enable/disable, and a screen-saver password option. The password option determines whether a password is required when exiting the screen saver.

Source page 22 — Auxiliary functions

The Auxiliary Functions screen displays additional controller information and settings.

The visible items include date/time, cumulative power-on time and background-light/screen brightness.

Power-on time refers to the accumulated time since the equipment was powered on.

Source page 23 — No legible text

No readable text was recovered from this scanned page. It appears to be blank or image-only.

Source page 24 — Level 1 / Level 2 over-temperature protection

This section explains how to configure the independent over-temperature protection device.

Level 1 protection

The visible instruction indicates that the high-temperature protection setting should be configured above the customer’s test temperature. The example describes a setting approximately 10°C above the customer’s maximum test temperature. “NC” appears to refer to the customer’s test-product protection temperature/limit; the exact abbreviation is not defined clearly in the scan.

Level 2 protection

The second-level protection is also configured above the customer’s maximum test temperature. The example again indicates a margin of approximately 10°C. The purpose is to protect the test product if the chamber temperature rises abnormally.

The page also describes the emergency stop switch: if an abnormal condition or loss of control occurs, press the emergency-stop switch to cut off the equipment’s main power.

The exact setting values shown in the examples are not completely legible. Set protection limits based on the test specimen’s allowable temperature and the equipment manufacturer’s instructions.

Source page 25 — Installation conditions and pre-test section

Cooling-water requirements — water-cooled models only

Water temperature: approximately 5°C to 28°C is stated as suitable for meeting the technical indicators; approximately 28°C to 33°C may permit normal operation but can reduce cooling performance.

Water pressure: the scan appears to specify approximately 0.3–0.6 MPa, with inlet/outlet differential pressure greater than approximately 0.2 MPa. Cooling-water circulation flow must meet the technical specification. The pipe diameter must be at least as large as the chamber cooling-water connection.

Environmental conditions

For air-cooled models, the stated ambient temperature range is approximately 25–35°C for normal operation and best performance. Ensure that the air inlet and outlet are unobstructed and that air can circulate freely.

For water-cooled models, the cooling-water supply must meet the requirements above. Relative humidity: below 85% RH. Atmospheric pressure: approximately 86–106 kPa. Avoid locations near strong electromagnetic radiation sources.

Provide a floor drain near the chamber for drainage.

Storage conditions

When the chamber is not in use, keep it in an environment above 0°C. If it must be stored below 0°C, drain residual water from water-cooled chambers and humidification systems to prevent freezing and damage to pipes.

Power supply

The distribution-box circuit breaker must be rated for the equipment’s maximum current. The power-cable conductor size must match the equipment’s total power.

The page lists supply options of 380 V ±10% or 220 V ±10%, at 50 Hz ±2%. It calls for an independent protective earth, with grounding resistance below 4 Ω, and a three-phase five-wire or single-phase three-wire AC supply as applicable. Do not connect neutral and protective earth together.

The OCR of the water-pressure, voltage and grounding figures is imperfect. Confirm the exact electrical and water requirements from the machine nameplate and the manufacturer before installation.

Source page 26 — Pre-test inspection and confirmation

Before starting a test, check the equipment condition and confirm that the required supply conditions are available.

Specimen placement and airflow

The specimen must not block the chamber’s air circulation. The manual gives the following placement guidance:

1. The specimen’s total volume (W × D × H) should not exceed approximately 20–35% of the effective working volume of the chamber; the scan appears to recommend about 29%.

2. The specimen’s windward/cross-sectional area should not exceed approximately 35–59% of the chamber working-section area; the scan appears to recommend about 35%.

3. Keep the specimen’s outer surface approximately 100–150 mm away from the chamber walls; the scan appears to recommend about 150 mm.

Specimen suitability and protection

Confirm that the specimen is not corrosive and will not corrode or damage the chamber materials. Check that its weight and dimensions are within the chamber’s limits and comply with the equipment technical specifications.

Confirm that the test parameters match the specimen’s test requirements and that the specimen will not be subjected to an unintended or excessive test.

Set protective measures in case of an unexpected chamber fault or shutdown. The manual describes an independent over-temperature protector, separate from the control system, to prevent the specimen from being damaged by excessive chamber temperature. It generally recommends setting this protection approximately 10°C above the test temperature.

For specimens that require powered operation during testing, the user should control the specimen’s power supply and arrange for it to disconnect automatically in an abnormal condition.

The next section is Fault Analysis and Troubleshooting.

Source page 27 — No legible text

No readable text was recovered from this scanned page. It appears to be blank or image-only.

Source page 28 — Fault analysis — heaters, humidifier and refrigeration system

The troubleshooting table lists common alarm conditions, possible causes and checks. The OCR of several alarm codes is uncertain; use the alarm displayed by the controller as the primary reference.

Heater over-current / heating fault

Possible causes include a heater circuit fault or excessive current. Check the heater wiring and electrical connections; inspect the heating element and associated control components.

Humidification over-current / humidification fault

Check the humidifier circuit and its electrical connections. The manual indicates that the humidification heater’s insulation resistance should be checked and that the relevant circuit breaker may need inspection.

Compressor / high-pressure fault

Possible causes include insufficient cooling-water flow or excessive cooling-water temperature (the scan refers to water temperature above approximately 30°C), a dirty air-cooled condenser, a condenser fan that is not operating, scale in a water-cooled condenser, blocked cooling-water pipes or poor heat dissipation. Check the cooling-water supply, condenser cleanliness and fan operation.

Compressor over-temperature / discharge-temperature fault

Check condenser airflow and cleanliness, fan operation, refrigerant charge, bypass-valve operation and temperature-sensor condition. The manual also refers to checking whether the hot-gas bypass solenoid valve is opening too far or is stuck.

Check whether the refrigerant is leaking or overcharged. Check whether cooling-water temperature is too low or too high, as applicable to the model.

The table’s exact alarm codes and some pressure/temperature thresholds are illegible. Do not use guessed values for diagnosis or adjustment.

Source page 29 — Fault analysis — oil pressure, water supply and humidification

Compressor electronic oil-pressure alarm

Check whether the condenser fan and cooling-water pressure are normal. The manual notes that excessive pressure difference or an oil-pressure protection device fault may trigger an alarm. Check whether the oil-pressure switch/solenoid is operating correctly, whether the compressor oil has darkened and needs replacement, and whether the compressor oil-return pipe is blocked.

Compressor cooling-water low-pressure / low-water alarm

Check that cooling water is available and that the water supply is clean and adequate.

Humidification heater over-temperature

Check whether the humidifier tank contains sufficient water and whether the water inlet is operating. Check whether the drain solenoid valve is closing properly. Inspect the humidifier over-temperature controller and sensor for damage.

Temperature fault / temperature does not reach the set value

Check the humidification water supply, water level and water-absorption condition. Check the water supply and drainage paths. Confirm that the ambient temperature and operating conditions are suitable. Inspect the solid-state relay and relevant control components.

Mechanical high-pressure / compressor pressure alarm

Check the cooling-water flow and temperature, condenser condition and refrigerant charge. Inspect the relevant compressor solenoid valves and bypass valves.

Compressor discharge-temperature high alarm

Check condenser cooling, fan operation (air-cooled models), water-side scaling (water-cooled models), cooling-water flow and temperature, refrigerant charge and the discharge-temperature sensor.

Source page 30 — Fault analysis — additional compressor alarms and alarm list

Additional checks for compressor discharge-temperature and suction-pressure alarms include checking condenser fans, condenser cleanliness, water-side scale, solenoid valves, refrigerant quantity and temperature/pressure sensors.

If the suction pressure is below the specified lower limit, check for refrigerant leakage, insufficient refrigerant, a valve that is not operating correctly, or an abnormal compressor condition.

Alarm troubleshooting summary

Alarm shown in manual	Possible cause	Recommended check / action
1 — Humidification alarm	Motor/pump fault or protection fault; possible blocked water pipe or no water in humidifier.	Check motor operation and protection device; clean the water pipe and confirm water supply.
2 — Heating alarm	Heating element or heater circuit fault.	Inspect the heating element and electrical circuit.
3 — Phase-sequence alarm	Incorrect phase sequence / phase wiring.	Have a qualified electrician verify and correct the phase sequence.
4 — Compressor protection	Compressor current too high or compressor protection activated.	Check compressor current and protection settings; remove the cause before resetting.
5 — Pressure alarm	Refrigerant/cooling-system pressure too high.	Check condenser fan operation and cooling system.
6 — Water-shortage alarm	Low water level or float switch fault.	Add water and inspect the float switch.

After correcting the fault, reset the relevant protection only when it is safe to do so and the cause has been identified.

Source page 31 — Refrigeration principle diagram

The page contains a refrigeration-system principle diagram. The drawing is model/configuration dependent and is identified in the source as an accessory/parts diagram.

The individual component tags and pipe labels are too small or blurred to translate reliably. Refer to the original high-resolution drawing and the machine-specific refrigeration schematic when servicing the system.

Visible diagram title: Refrigeration Principle Diagram (configuration depends on machine model).

Source page 32 — Equipment / electrical schematic

This page contains a machine schematic/electrical diagram. The scan is too faint to read the individual terminal numbers, wire labels and component references reliably.

Do not use this translated text as a wiring reference. Use the original machine-specific electrical drawing and have electrical work performed by qualified personnel.

Source page 33 — Wiring / circuit diagram

This page contains a further wiring or control-circuit diagram. Individual labels, terminal references and wiring connections are not sufficiently legible in the scan for safe transcription.

Use the original drawing supplied with the equipment for wiring, troubleshooting and component replacement.

Source page 34 — Quarterly maintenance checklist — first part

The page is a periodic maintenance inspection form. It contains columns for maintenance item, maintenance content/method, inspection result/status, and remarks or corrective-action records.

The visible checklist items include:

• Inspect the compressor operating condition and record abnormal noise or vibration.

• Inspect the fan and fan motor for normal operation.

• Check compressor operating pressure and operating conditions; record the readings.

• Check piping, joints and connections for leakage or abnormal condition.

• Check the humidification water supply, water inlet connection and drain piping.

• Open the water inlet valve and confirm that water enters the humidifier tank; confirm that the water-level control operates.

• Check humidifier heating operation and confirm that the water level is adequate during operation.

• Check the over-temperature protection and heater protection functions.

The form provides checkboxes for “normal / requires attention” and space to record corrective actions.

Source page 35 — Quarterly maintenance checklist — second part

The maintenance form continues with inspection of the refrigeration, ventilation, electrical and control systems.

Visible checklist items include:

• Inspect condenser/heat-exchanger airflow and clean accumulated dust or debris.

• Check the refrigeration system and related piping/valves for abnormal conditions.

• Inspect the fresh-air/ventilation system; confirm that the fan operates and that air can circulate.

• Check the compressor and system operating condition.

• Check the internal chamber and external equipment condition; clean as required.

• Inspect electrical wiring and terminals for looseness, damage or abnormal heating.

• Record compressor operating current and humidifier/heater current, and compare with the normal operating values.

The checklist includes fields for the responsible person, date, inspection result and corrective-action notes. Some individual row labels are blurred.

Source page 36 — No legible text

No readable text was recovered from this scanned page. It appears to be blank or image-only.

Source page 37 — Routine maintenance schedule and instructions

During maintenance, switch off the machine’s main power supply before carrying out work. Perform periodic maintenance to help ensure normal operation.

Maintenance item	Recommended interval	Maintenance method
Internal and external surfaces / machine body	Every 7 days	Clean and remove dust/debris.
Power cable, power switch and protective earth	Every 30 days	Inspect and tighten as necessary.

The manual states that 24 hours of operation counts as one day. This page is provided as a maintenance reference.

1. Internal and external surfaces

1) Clean dust, dirt and stains from the internal and external surfaces and components.

2) Wipe with a soft cloth to avoid scratching the surfaces.

3) Do not use corrosive cleaning agents.

2. Power cable, power switch and grounding

Periodically inspect the wiring terminals of the power cable and main power switch, as well as the protective-earth connection. If a connection is loose, tighten it with a screwdriver by turning clockwise.

Source page 38 — Quality warranty certificate

Warranty period shown: 22 May 2026 to 21 May 2027.

The warranty card states a 12-month warranty. During the warranty period, faults caused by product quality problems may be repaired free of charge, subject to the stated exclusions.

Situations excluded from warranty

1. Damage or failure caused by natural disasters or other force-majeure events.

2. Damage caused by incorrect use or modification of the machine.

3. Failure caused by moving or transporting the machine after installation.

4. Damage caused by improper operation or use.

5. Damage caused by assembly or repair by another manufacturer/third party.

6. Failure caused by repair or disassembly by personnel who are not authorized service staff of the manufacturer.

If the warranty card is lost, the warranty period is stated to run for 12 months from the date the product is delivered to the customer.

Source page 39 — Warranty card

The final page is a warranty/service record card with fields for customer and equipment information, service records and dates. The printed field labels are partly illegible.

The manual instructs the customer to present the record card when requesting service. If the card is lost, the warranty period is calculated from the delivery date.

For equipment faults, contact the manufacturer in writing and confirm by telephone. The manufacturer states that it will arrange service promptly.

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